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Project Title:	The Mouse as an Animal Model for Human Bone Loss During Spaceflight		
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Contact Monitor:		**Contact Phone:**	
Contact Email:			
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Flight Assignment:	NOTE: Received NCE through 7/31/2008 per A. Chu (4/25/2007) No-cost extension until 7/31/2007, per PI (8/14/06)		
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Task Description:	Spaceflight (skeletal unloading) is associated with loss of bone mass. The mechanisms for this loss are not known. Our objective is to develope an animal model that mimics human bone loss during spaceflight. The goal is to use such a model to study how unloading affects bone cell function and mass. We have chosen to work with the mouse because it is small and in other studies has ben shown to mimic the human with respect to aging, and responsiveness to various endocrine factors. Because there are numerous mouse strains and each behaves differently, we will evaluate three strains. We will use the hindlimb unloading model to induce skeletal unloading and follow the mice over time. Bone formation, mass and structure will be assessed. To determine whether the mouse is a suitable model to study adult human bone loss during spaceflight and recovery after re-ambulation. It is essential to have an animal model with which to study the loss of bone that occurs with spaceflight. Without this it is not possible to determine the mechanisms and thus devise means of reducing bone loss and restoring bone after flight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Disuse osteoporosis is associated with sedentary lifestyles as well as spaceflight and contibutes to fractures of the hip and spine. Our studies are directly applicable to disuse osteoporosis. Understanding how bone responds to loading through animal models will provide a means of defining mechanisms.
Task Progress:	Our goal was to determine whether the mouse is a suitable model to study adult human bone loss during spaceflight, and to examine the effects of reloading in animals treated with IGF-I. MicroCT analysis, bone histomorphometry, and cell culture were used to assess bone structure and makeup, bone formation and osteoprogenitor number and osteoclastogenesis. Over the course of our work we completed the aims of our grant and in addition conducted several other related studies. Our study examined two different strains of mice, the DBA and C57B/6 (C57). Mean body weights did not show any significant difference between hindlimb unload versus control mice at the start of the unloading experiment. Although body weights trended downward at day 7, the difference was not statistically significant and body weights in both strains stabilized after 14 days of suspension, ultimately rebounding to basal (pre-suspension) body weight by day 28 of suspension. There was no significant difference between control C57 and DBA body weights. DBA mice had a lower bone volume than C57 mice. With unloading both lost bone proportionately. Micro CT analysis from both strains showed no significant difference in bone fractional volume (BV/TV) at the 1-week timepoint. After 2 weeks of suspension unloaded C57 mice had 30% lower BV/TV than loaded controls, while DBA mice still showed no significant decrease in BV/TV. Hindlimb unload mice of both strains showed a decrease in BV/TV at the 4-week timepoint, with C57 unload mice showing a larger decrease than DBA (35%, p=0.001 and 32%, p=0.039 respectively). DBA hindlimb unloaded mice experienced no significant change in total bone mineral density with time in suspension, at any of the 1, 2 or 4 week timepoints. C57BL6 hindlimb unloaded mice, in contrast, experienced a 26% decrease with 2 weeks of unloading and a 39% decrease with 4 weeks of unloading. No significant segmented bone mineral density changes were observed, except a 3.5% decrease in the C57BL6 4-week unload group. Trabecular number data indicate no change with unloading in DBA mice at 1, 2 and 4 week timepoints. In C57BL6 mice, however, there is a 10% decrease in trabecular number with 2 weeks of suspension. None of the other timepoints produced any significant changes. In both strains of mice, trabecular thickness decreased continuously with unloading. With 1 week of unloading, neither mouse strain saw a significant change in Tb.Th, but at the 2-week timepoint, a 10% decrease was observed in both the C57BL6 and the DBA/2 unloaded mice. This decrease continued in the 4-week timepoint with a 21% and 15.7% decrease in the C57 and DBA unloaded mice respectively. Structure model index data reveal a lower SMI in C57 versus DBA, indicating a more plate-like structure in C57 trabeculae (data not shown). Bone formation at the tibio-fibular junction decreased but did not reach significance. Bone mineral density both total and segmented (density of the bone mineral alone) decreased in the C57 but not in the DBA. Osteoprogenitor number decreased initially and then gradually returned to the control level in both strains. Osteoclastogenesis increased to the same degree in both strains and then gradually returned to normal. Our data indicate that both strains respond to unloading but with important differences. In both strains osteoblast formation decreases and osteoclast formation increases. These changes are consistent with the trend for bone formation to decrease with time of unloading. That bone mineral density decreased in the C57 but not in the DBA suggests that the response of these two strains are not identical. After 1 week of skeletal unloading, osteoblast cultures from both mouse strains showed significantly decreased calcium nodule formation when compared to those cultured from 1 week loaded control animals. Osteoblast cultures isolated from 1 week unloaded C57 mice exhibited a 70% decrease in alizerin red (AR) mineralized nodule formation when compared to those isolated from normally loaded control C57 mice. Cultures isolated from DBA mice at this same timepoint showed a 60% decrease. By 2 and 4-week time points, cultures isolated from unloaded animals in both strains, had half as many nodules as normally loaded animals (p=0.001). There was no significant change in nodule formation over time of suspension in unloaded animals, although nodule formation in unloaded animals trended upward with time in suspension. Osteoclasts cultured from 1-week unloaded mice showed a two-fold increase in total osteoclast number when compared to those isolated from 1 week normally loaded control animals. Results are nearly identical in both C57 and DBA mice. This difference in total number of osteoclasts was reduced after 2 weeks of suspension, and was nearly non-existent at the 4-week. When osteoclasts in each culture were categorized by size, a more striking result was observed. The large osteoclast (osteoclasts containing >50 nuclei) numbers revealed the most pronounced difference between load and unload groups at the 1-week time point. At the 1-week time point, osteoclast cell cultures isolated from unloaded C57 mice contained 3.5 times the number of large osteoclasts isolated from normally loaded animals. Osteoclast cell cultures isolated from 1 week suspended DBA mice produced 4 times more large osteoclasts than normally loaded 1 week cultures. These differences between large osteoclasts in normally loaded versus unloaded groups became indistinguishable after 4 weeks of suspension in both C57 and DBA mouse strains. The changes we observed with unloading in the mouse were very similar to those observed in humans during un-weighting or disuse. Osteoprogenitor number and bone formation decreased and importantly osteoclast progenitor number and bone resorption increased. These findings strongly support the use of the mouse as an animal model to study adult human bone loss during space flight. Interestingly bone resorption increased dramatically in our mice but in previous studies it has been reported that resorption in the rat contributes much less to bone loss during unloading. Although rats have been used extensively in space flight experimentation, studies have shown that although bone formation is decreased in ground based models, decreased bone formation during space flight is minimal or non-existent of bone. Resorption in the rat is also difficult to demonstrate in ground based studies. Thus our findings suggest that the

	mouse is likely to be a better animal model for human bone during space flight. We also examined the effects of reloading after a period of unloading in animals treated with IGF-I. Our studies show that reloading increases bone formation and gradually restores bone mass. The process is, however, slow and complete restoration of bone may never occur. To determine if we could increase bone recovery and reduce fracture risk, we treated animals with IGF-I during the recovery period. Our findings suggest that IGF-I treatment speeds bone recovery by stimulating bone formation. Thus IGF-I administration may be of benefit in restoring bone in astronauts after long duration space missions, and in treating patients following a period of immobilization when such patients are returned to normal weight bearing.
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